

SCI225 · PATHOPHYSIOLOGY · WEEK 8 · NIGHTINGALE COLLEGE

The Hematologic System

Blood components, the anemias, sickle cell disease, and the bleeding-versus-clotting framework that shapes nursing care — built around the lab signatures you will use at the bedside.

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Chapters 27 – 29

Structure · Alterations · Clinical

21 topics + cheat sheet

Structure & Function

SECTION 1

Blood components, erythrocyte design, hemoglobin, erythropoiesis, and hemostasis — the foundation for every disorder.

Blood is about **55% plasma** (the liquid) and **45% formed elements** (the cells and fragments). Each component has one job — and each one, when it fails, produces a predictable disease pattern.

Component	What it is	Primary function	When it fails...
Plasma	92% water + albumin, globulins, fibrinogen, electrolytes	Transport, oncotic pressure, clotting, immunity	Low albumin → edema
Erythrocytes (RBCs)	Biconcave, anucleate; packed with hemoglobin	Carry O ₂ and CO ₂	Anemia → tissue hypoxia, fatigue
Leukocytes (WBCs)	Neutrophils, lymphocytes, monocytes, eosinophils, basophils	Immune defense & inflammation	Leukopenia → infection risk
Platelets (thrombocytes)	Cell fragments from megakaryocytes	Hemostasis — the first plug	Thrombocytopenia → bleeding

Plasma proteins — why albumin matters

Albumin is the most abundant plasma protein and the main driver of **plasma oncotic (colloid osmotic) pressure** — the “pull” that keeps fluid inside the vessels. Albumin is made by the liver.

Key idea — the liver-disease connection: In severe liver disease the liver makes less albumin. Plasma oncotic pressure falls, so fluid leaks out of the capillaries into the interstitium → **generalized edema**. (Not increased permeability, not venous pressure — the primary problem is low oncotic pull.)

Match each problem to the right cell line: oxygen problems = RBCs; infection problems = WBCs; bleeding problems = platelets; fluid-balance problems =

plasma proteins. This single matching habit answers a surprising number of exam items.

PRACTICE

Q1. A patient with severe liver disease develops generalized edema. Which physiologic change best accounts for this finding?

- A. Decreased albumin synthesis reducing plasma oncotic pressure
- B. Increased venous pressure increasing capillary hydrostatic pressure
- C. Decreased lymphatic drainage reducing interstitial fluid removal
- D. Increased capillary permeability increasing outward fluid movement

A is correct. A failing liver synthesizes less albumin, lowering plasma oncotic pressure so fluid shifts into the interstitial space. B and D can cause edema in other settings (heart failure, inflammation) but are not the primary mechanism in liver disease; C describes lymphedema.

PRACTICE

Q2. Which function is primarily associated with platelets in hemostasis?

- A. Supporting vessel constriction and forming temporary plugs
- B. Carrying respiratory gases and delivering oxygen
- C. Producing immune proteins and targeting pathogens
- D. Maintaining plasma pressure and supporting fluid balance

A is correct. Platelets form the primary plug and support clotting. B is the RBC role, C belongs to lymphocytes/leukocytes, and D is the role of plasma proteins such as albumin.

The red blood cell is a specialist — every structural feature exists to move oxygen efficiently.

Why the RBC is shaped the way it is

- **Biconcave disc** — maximizes surface area for gas diffusion and lets the cell deform to squeeze through narrow capillaries.
- **No nucleus** — frees up room for more hemoglobin; the trade-off is that a mature RBC **cannot divide**.
- **No mitochondria** — the RBC runs on anaerobic glycolysis, so it will not burn the very oxygen it is carrying.
- **Lifespan ~120 days** — aged cells are removed by macrophages in the **spleen** (the reticuloendothelial system).
- ~270 million hemoglobin molecules per cell, each able to bind up to **4 O₂**.

Hemoglobin structure

Adult hemoglobin (HbA) has **four globin chains** (2 alpha, 2 beta). Each chain holds one **heme** group with one iron atom in the **ferrous (Fe²⁺)** state, and only ferrous iron can bind oxygen. Cooperative binding gives the familiar S-shaped (sigmoidal) oxygen–dissociation curve.

What can go wrong	Mechanism	Result
Carbon monoxide	Binds Hb ~250× more tightly than O ₂ → carboxyhemoglobin	O ₂ cannot board; tissue hypoxia despite “normal” SpO ₂
Methemoglobinemia	Iron oxidized Fe ²⁺ → Fe³⁺ (ferric)	Ferric iron cannot bind O ₂
Sickle Hb (HbS)	Point mutation; polymerizes when deoxygenated	Rigid, sickled cells
Thalassemia	Reduced globin-chain production	Microcytic anemia

Pearl — CO vs. methemoglobin: Carbon monoxide *occupies* the binding site (carboxyhemoglobin). Methemoglobinemia *changes the iron* to the ferric form so

it cannot bind at all. Both lower oxygen delivery, but by different mechanisms — a favorite exam distinction.

PRACTICE

Q1. Which characteristic of erythrocytes supports efficient oxygen exchange?

- A. Biconcave shape
- B. Nuclear content
- C. Mitochondrial activity
- D. Cell division

A is correct. The biconcave shape increases surface area and diffusion efficiency. Mature RBCs deliberately *lack* a nucleus and mitochondria and cannot divide — all of which maximize hemoglobin capacity.

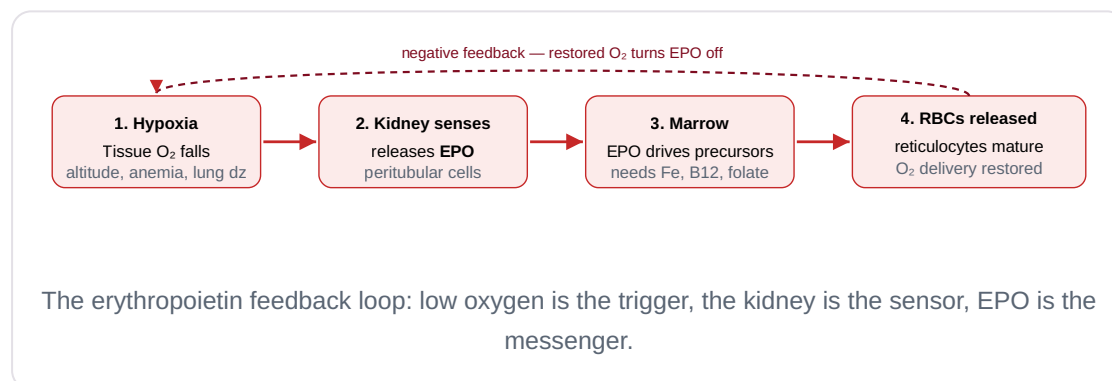
PRACTICE

Q2. A patient exposed to carbon monoxide shows decreased oxygen delivery. Which mechanism explains this effect?

- A. Carbon monoxide binds hemoglobin with high affinity
- B. Carbon monoxide oxidizes hemoglobin iron to the ferric form
- C. Carbon monoxide lowers oxygen release from hemoglobin
- D. Carbon monoxide blocks oxygen diffusion in alveoli

A is correct. CO binds hemoglobin far more strongly than oxygen, forming carboxyhemoglobin and occupying the binding sites. Oxidation of iron to the ferric form (B) describes methemoglobinemia, a different problem.

The body keeps oxygen delivery steady through a tight feedback loop centered on the kidney and the hormone **erythropoietin (EPO)**.



Key idea: Low oxygen — not low blood, not low iron — is the trigger.

Hypoxia (high altitude, chronic lung disease, anemia, hemorrhage, after blood donation) makes the kidney release EPO, which stimulates the marrow. This explains the high hematocrit of high-altitude dwellers and the reticulocyte rise after bleeding.

Required raw materials

Even with plenty of EPO, the marrow needs **iron, vitamin B12, folate, and protein**. Missing any one disrupts production — the basis for the nutritional anemias in Section 2.

Clinical link: Diseased kidneys make less EPO → the **anemia of chronic kidney disease**, treated with synthetic EPO. A premature newborn's immature kidneys also under-produce EPO → anemia of prematurity.

PRACTICE

Q1. A patient living at high altitude develops increased red blood cell production. Which mechanism drives this change?

- A. Renal hypoxia triggers erythropoietin secretion
- B. Hemoconcentration raises measured hematocrit values
- C. Hyperventilation lowers arterial carbon dioxide

D. Sympathetic output raises heart rate response

A is correct. Low oxygen tension causes renal hypoxia, stimulating EPO release and true RBC production. B raises the *measured* hematocrit by concentrating the blood but makes no new cells; C and D are other altitude responses, not the erythropoietic driver.

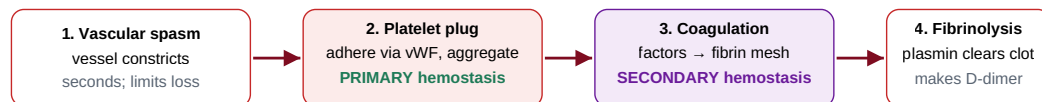
PRACTICE

Q2. After blood donation, which mechanism restores the red blood cell count?

- A. Kidney release of erythropoietin increases marrow production
- B. Splenic removal of erythrocytes increases blood oxygenation
- C. Platelet activation in vessels increases clotting efficiency
- D. Plasma protein synthesis in the liver increases fluid retention

A is correct. The drop in oxygen-carrying capacity is sensed as hypoxia, the kidney releases EPO, and the marrow makes new RBCs. The spleen *removes* cells, platelets handle hemostasis, and plasma proteins handle fluid balance — none replaces lost RBCs.

Hemostasis stops bleeding in an orderly sequence. Splitting it into **primary** (platelets) and **secondary** (clotting factors) is the key to reading the lab panel later.



Four steps: spasm → platelet plug (primary) → fibrin clot (secondary) → controlled breakdown.

Primary hemostasis — the platelet plug

Platelets **adhere** to exposed subendothelial collagen using **von Willebrand factor (vWF)** as the bridge, then **activate** and release ADP and thromboxane A_2 to recruit more platelets, and finally **aggregate** as **GPIIb/IIIa** receptors bind fibrinogen between platelets. The result is a soft “primary plug.” Defects here cause **mucosal bleeding, petechiae, and a prolonged bleeding time**.

Secondary hemostasis — the fibrin mesh

The clotting cascade converts **fibrinogen into fibrin**, which stabilizes the platelet plug into a firm clot. Defects here (clotting-factor problems) cause **deeper bleeding** — into joints and muscles.

Pearl — the on/off signals: Intact endothelium is **antithrombotic** (nitric oxide and prostacyclin keep platelets calm). Injury exposes collagen and tissue factor, flipping the surface to **prothrombotic**. Nitric oxide therefore *inhibits* clotting — a classic distractor.

PRACTICE

Q1. Which event occurs during secondary hemostasis?

- A. Conversion of fibrinogen into fibrin strands
- B. Constriction of vessels to reduce blood flow
- C. Adhesion of platelets to exposed collagen

D. Release of nitric oxide from vessel walls

A is correct. Secondary hemostasis is the cascade that produces fibrin to stabilize the clot. B is the initial vascular response, C is primary hemostasis, and D actually *inhibits* clotting.

PRACTICE

Q2 (Select all). Which events directly strengthen platelet plug formation at the injury site?

- A. ADP release recruits additional platelets locally
- B. Thromboxane A₂ promotes vasoconstriction and activation
- C. Nitric oxide exposure increases platelet aggregation
- D. GPIIb/IIIa activation allows fibrinogen bridging between platelets
- E. Tissue plasminogen activator preserves the early plug

A, B, and D are correct. ADP recruits platelets, thromboxane A₂ drives vasoconstriction and activation, and GPIIb/IIIa binds fibrinogen to bridge platelets. C is wrong — nitric oxide *inhibits* activation; E is wrong — tPA promotes fibrinolysis, breaking the plug down.



You do not need to memorize every factor — you need to know **which pathway each lab measures**. Four numbers (PT, aPTT, platelets, D-dimer) decode most coagulopathies.

Pathway / test	Trigger & key factors	Lab & what it monitors
Intrinsic	Damage to blood; factors XII, XI, IX, VIII	aPTT · monitors heparin
Extrinsic	Tissue injury; factor VII + tissue factor	PT / INR · monitors warfarin
Common	Factors X, V, II (prothrombin), I (fibrinogen)	Both PT <i>and</i> aPTT prolonged
Platelet count	Primary hemostasis — adhesion & plug	Low → bleeding (petechiae, purpura)
D-dimer	Fibrin breakdown product	Elevated in DVT, PE, DIC

MEMORY HOOK

PiII = intrinsic → **aPTT** · **PeI = extrinsic** → **PT**. Also: **warfarin** lengthens the **PT** (both have a long “a” sound), **heparin** lengthens the **aPTT**.

Reading the panel: Isolated long **aPTT** → intrinsic (hemophilia, heparin, vWD). Isolated long **PT** → extrinsic (warfarin, vitamin K deficiency, early liver disease). **Both** long → common pathway or liver disease. Low **platelets** with normal PT/aPTT → a platelet problem (ITP). This framework returns in Topic 15.

PRACTICE

Q1. A patient on heparin needs monitoring. Which test reflects heparin’s effect on the intrinsic pathway?

- A. aPTT
- B. PT / INR

C. Platelet count

D. D-dimer

A is correct. aPTT monitors the intrinsic pathway and heparin therapy. PT/INR monitors the extrinsic pathway and warfarin; platelet count and D-dimer measure different things entirely.

PRACTICE

Q2. A patient's D-dimer is markedly elevated. What does this finding indicate?

A. Active fibrin breakdown, as in DVT, PE, or DIC

B. A deficiency of vitamin K

C. Reduced platelet production in the marrow

D. Impaired oxygen transport by hemoglobin

A is correct. D-dimer is a fibrin-degradation product; a high level signals that clot is forming and being broken down somewhere — classically DVT, PE, or DIC.



White blood cells defend the body. For Week 8 you need their core jobs and what happens when their numbers fall.

Leukocyte	Main role
Neutrophils	First responders; phagocytosis of bacteria (most numerous WBC)
Lymphocytes	Adaptive immunity — B cells make antibodies, T cells kill/coordinate
Monocytes / macrophages	Phagocytosis + antigen presentation; recycle old RBCs in the spleen
Eosinophils	Parasites and allergic responses
Basophils	Release histamine; drive inflammation/allergy

Key defense actions

- **Phagocytosis** — engulfing and digesting a pathogen; the action that *directly eliminates* microorganisms.
- **Chemotaxis** — movement *toward* the infection (guides cells in, but does not kill).
- **Antigen presentation** — activates other immune cells (support, not direct killing).
- **Degranulation** — releases mediators that amplify inflammation.

Key idea — leukopenia: A low WBC count weakens the body's ability to recognize and eliminate pathogens, so **infection risk rises**. This is the central danger in chemotherapy, leukemia (marrow crowding), and radiation-induced marrow suppression.

PRACTICE

Q1. Which leukocyte function is most directly responsible for eliminating invading microorganisms?

- A. Phagocytosis
- B. Degranulation
- C. Antigen presentation
- D. Chemotaxis

A is correct. Phagocytosis directly engulfs and digests the pathogen. Degranulation amplifies inflammation, antigen presentation activates other cells, and chemotaxis only guides cells to the site.

PRACTICE

Q2. A patient receiving chemotherapy develops a low white blood cell count. Which mechanism explains the increased risk of infection?

- A. Impaired leukocyte defense
- B. Impaired erythrocyte transport
- C. Impaired platelet function
- D. Impaired plasma protein balance

A is correct. Leukopenia means fewer white cells to defend against pathogens. Erythrocytes (B) carry oxygen, platelets (C) handle clotting, and plasma proteins (D) handle fluid balance — none explains infection risk.

SECTION 2

Alterations of Hematologic Function

The anemias, sickle cell disease, polycythemia, and the bleeding-versus-clotting framework — each anchored to its lab signature.

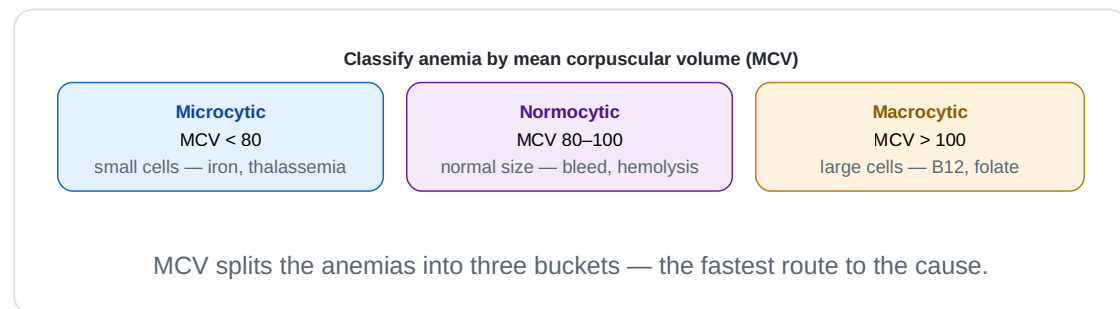
Anemia = reduced oxygen-carrying capacity of the blood, defined as **Hb < 13 g/dL (men)** or **< 12 g/dL (women)**, or reduced RBC mass. It is a *sign* of an underlying problem, not a diagnosis by itself.

General symptoms (any anemia)

Fatigue, pallor, tachycardia, dyspnea on exertion, and cold extremities — all stem from one root cause: **less oxygen reaching the tissues**.

Two ways to classify — learn both

By size (MCV) — the most useful first cut on a lab report:



By mechanism — (1) decreased production (nutrient, EPO, or marrow problems), (2) increased destruction (hemolysis), or (3) blood loss.

How the body compensates

When oxygen delivery drops, the body responds quickly and slowly:

- **Fast:** sympathetic drive raises **heart rate** and cardiac output; respiratory rate rises; hemoglobin unloads more oxygen to tissues.
- **Slow:** the kidney releases **EPO**, pushing the marrow to raise the **reticulocyte** count.
- Fewer RBCs also lower blood **viscosity**, so blood flows faster — another way to move oxygen.

Key idea: The fatigue and dyspnea of anemia come from **reduced oxygen delivery to tissues** — not lost plasma volume, not a diffusion problem in the lungs. Compensation (tachycardia, tachypnea, EPO, increased reticulocytes) is the body trying to fix that single deficit.

PRACTICE

Q1. A patient with anemia reports persistent fatigue. Which pathophysiologic change explains this symptom?

- A. Reduced oxygen delivery to tissues
- B. Reduced carbon dioxide removal
- C. Reduced oncotic pressure
- D. Reduced platelet activity

A is correct. Fewer functional RBCs lower oxygen-carrying capacity, causing tissue hypoxia and fatigue. CO₂ removal, oncotic pressure, and platelet activity are unrelated to oxygen transport.

PRACTICE

Q2. Which compensatory response helps maintain tissue oxygen delivery in a patient with anemia?

- A. Increased heart rate
- B. Increased platelet count
- C. Decreased respiratory rate
- D. Decreased plasma volume

A is correct. The body raises cardiac output (heart rate) to compensate for reduced oxygen delivery. Respiratory rate *increases* rather than decreases, and platelets/plasma volume are not compensatory for oxygen transport.

Small, often pale (hypochromic) cells. The unifying theme is **too little hemoglobin per cell** — usually an iron or globin problem.

Type	Cause	Key labs / clues
Iron-deficiency	Blood loss, poor intake, malabsorption, pregnancy	↓ ferritin, ↓ serum iron, ↑ TIBC; pica, spoon nails
Thalassemia	Inherited reduction in globin-chain production	Normal iron studies, target cells, family history
Anemia of chronic disease/inflammation	Inflammation traps iron in macrophages (hepcidin)	↑ ferritin, ↓ serum iron, ↓ TIBC
Sideroblastic	Defective heme synthesis	Ringed sideroblasts on marrow biopsy

The hepcidin story (anemia of chronic inflammation): Inflammatory cytokines raise **hepcidin**, which degrades ferroportin and **traps iron inside macrophages and gut cells**. Serum iron falls even though total-body iron (ferritin) is normal or high — iron is present but *unavailable*. This is exactly Elena's case study: low serum iron, low transferrin saturation, *high* ferritin.

Ferritin is the discriminator: Iron-deficiency → ferritin **LOW** (stores empty). Anemia of chronic disease → ferritin **HIGH** (stores full but locked away). Same small cells, opposite ferritin.

PRACTICE

Q1. A patient has microcytic, hypochromic erythrocytes. Which mechanism best explains this finding?

- A. Decreased hemoglobin synthesis leading to smaller, pale erythrocytes
- B. Increased erythrocyte destruction leading to larger, fragile cells
- C. Increased plasma volume diluting normal erythrocytes
- D. Decreased DNA synthesis producing larger, immature cells

A is correct. Iron deficiency reduces hemoglobin production, producing small, pale cells. Decreased DNA synthesis (D) makes *macrocytic* cells; hemolysis and dilution do not create microcytosis.

PRACTICE

Q2. A patient with chronic inflammation develops anemia despite adequate iron stores. Which alteration explains this finding?

- A. Increased hepcidin activity trapping iron in storage sites
- B. Decreased iron availability despite normal total body iron
- C. Increased transferrin activity delivering iron to marrow
- D. Decreased cytokine activity reducing marrow suppression

A is correct. Chronic inflammation raises hepcidin, which traps iron in storage and limits iron for erythropoiesis despite adequate stores. B describes the *effect* rather than the specific mechanism; C and D would *improve* erythropoiesis.

Large, immature cells. The shared defect is **impaired DNA synthesis**: the nucleus matures too slowly while the cytoplasm keeps growing, so cells get big before they can divide (megaloblastic anemia).

Cause	Why	Distinguishing feature
Vitamin B12 deficiency	Pernicious anemia (autoimmune loss of intrinsic factor), strict vegan diet, ileal disease	Neurologic signs: paresthesias, ataxia
Folate deficiency	Alcoholism, malnutrition, pregnancy, methotrexate	No neurologic signs

Shared lab clues: **elevated MCV, hypersegmented neutrophils**, and a low B12 or folate level.

Pearl — the deciding question: Both cause big cells. Only **B12 deficiency causes neurologic symptoms** (the myelin needs B12). Numbness, tingling, and unsteady gait point to B12, not folate.

Mechanism in one line: No B12/folate → no DNA → nucleus lags behind cytoplasm → large, immature RBCs. Contrast with iron deficiency, where the problem is too little *hemoglobin* (small cells).

PRACTICE

Q1. A patient with vitamin B12 deficiency develops large, immature red blood cells. Which mechanism explains this finding?

- A. Reduced DNA synthesis delaying nuclear maturation
- B. Increased iron absorption increasing hemoglobin production
- C. Reduced membrane flexibility increasing cell destruction
- D. Increased erythropoietin accelerating cell release

A is correct. B12 is required for DNA synthesis; without it, nuclear maturation and division lag while the cell keeps growing, producing large, immature RBCs. The other options describe unrelated processes.

PRACTICE

Q2. Which clinical feature best distinguishes vitamin B12 deficiency from folate deficiency?

- A. Neurologic symptoms such as paresthesias and ataxia
- B. A markedly elevated MCV
- C. The presence of hypersegmented neutrophils
- D. Fatigue and pallor

A is correct. Neurologic involvement is unique to B12 deficiency. Elevated MCV, hypersegmented neutrophils, fatigue, and pallor occur in *both* and cannot separate them.

Cells are normal-sized; the problem is loss, destruction, or under-production — not a size defect.

Type	Mechanism	Clinical clue
Acute blood loss	Hemorrhage; Hb falls after volume is replaced	Trauma, GI bleed, postpartum
Hemolytic	Premature RBC destruction	↑ reticulocytes, ↑ LDH, ↑ indirect bilirubin, ↓ haptoglobin
Aplastic	Bone marrow failure → pancytopenia	Drugs, radiation, idiopathic
Chronic kidney disease	↓ EPO production	Treat with synthetic EPO

Hemolytic anemia — recognize the signature

When RBCs are destroyed faster than they are made, heme breakdown releases **indirect (unconjugated) bilirubin** → **jaundice**, and the marrow ramps up **reticulocytes**. Destruction can be *extravascular* (spleen traps abnormal cells — e.g., spherocytes) or intravascular.

Key idea — the spleen as filter: The spleen removes abnormal/aged RBCs. **After splenectomy**, defective cells persist in circulation longer because that clearance is gone. In hereditary spherocytosis, rigid spherocytes are trapped in the splenic cords and destroyed — classic **extravascular hemolysis** (Iris's case study).

Aplastic vs. hemolytic: Aplastic anemia = marrow makes too *few* of everything (pancytopenia, **low** reticulocytes). Hemolytic = marrow works overtime (**high** reticulocytes) but cells are destroyed. Radiation and certain drugs cause the aplastic/pancytopenic picture.

PRACTICE

Q1. A patient presents with anemia, jaundice, and elevated indirect bilirubin. Which alteration explains this condition?

- A. Accelerated breakdown of erythrocytes
- B. Reduced formation of leukocytes
- C. Accelerated formation of platelets
- D. Reduced synthesis of plasma proteins

A is correct. Hemolytic anemia destroys RBCs faster than they are produced; heme breakdown raises indirect bilirubin, producing jaundice. Leukocytes, platelets, and plasma proteins do not explain this triad.

PRACTICE

Q2. A patient with radiation exposure develops pancytopenia. Which mechanism explains this condition?

- A. Damage to bone marrow affecting production of all blood cells
- B. Altered erythrocyte membrane affecting oxygen transport
- C. Disrupted platelet activation affecting clot formation
- D. Modified leukocyte signaling affecting immune response

A is correct. Radiation damages the marrow, reducing RBCs, WBCs, *and* platelets together (pancytopenia). The other options each describe only one cell line and cannot explain a drop in all three.

An inherited hemoglobinopathy with a single, memorable molecular cause and a high-yield set of crisis triggers.

Pathophysiology

- Single **point mutation**: glutamic acid → valine at position 6 of the **beta-globin** chain.
- The abnormal hemoglobin (**HbS**) **polymerizes when deoxygenated**, deforming the cell into a rigid sickle shape.
- Sickled cells **occlude the microvasculature** → ischemia and pain (vaso-occlusion).
- RBC lifespan shortens to ~10–20 days → chronic **hemolytic anemia**.

Crisis triggers — anything that promotes deoxygenation or sludging

Hypoxia, dehydration, infection, cold, acidosis, and stress. Recognizing these guides both prevention and the “what made this worse” exam question.

Why it hurts: rigid sickled cells physically block small vessels, cutting off blood flow → tissue ischemia → the severe pain of a **vaso-occlusive crisis** (often bones, chest, abdomen). **Acute chest syndrome** (fever + chest pain + new infiltrate) is a medical emergency.

PRACTICE

Q1. A child with sickle cell disease develops recurrent pain episodes. Which pathophysiologic mechanism explains this finding?

- A. Rigid red cells obstruct vessels and reduce tissue perfusion
- B. Fewer red cells reduce viscosity and increase tissue perfusion
- C. Large red cells carry more oxygen and lower tissue ischemia
- D. Fragile red cells raise plasma volume and improve circulation

A is correct. Sickled, rigid cells block small vessels, causing ischemia and recurrent vaso-occlusive pain. The other options describe improved flow or oxygenation, the opposite of what occurs.

PRACTICE

Q2. Which condition is most likely to trigger a sickle cell vaso-occlusive crisis?

- A. Dehydration and hypoxia
- B. Overhydration and rest
- C. Warm ambient temperature
- D. Mild alkalosis

A is correct. Dehydration and hypoxia (along with infection, cold, acidosis, and stress) promote HbS polymerization and sickling. Hydration, warmth, and rest help *prevent* crises.

The mirror image of anemia: an **excess** of red cells that thickens the blood and raises clot risk.

Type	Mechanism	EPO level
Primary (polycythemia vera)	JAK2 mutation → marrow autonomously overproduces RBCs (often WBCs & platelets too)	↓ EPO (suppressed)
Secondary	Appropriate response to chronic hypoxia — high altitude, COPD, heart disease, smoking	↑ EPO (elevated)

Symptoms: ruddy complexion, headache, **pruritus after a warm shower** (classic for PV), and hyperviscosity → **thrombosis risk**. **Treatment:** phlebotomy, low-dose aspirin, and treating the underlying hypoxia in secondary cases.

Key idea — why clots: more red cells → **higher viscosity** → sluggish flow → thrombosis. The EPO level separates the two types: **low** EPO = primary (the marrow is the problem); **high** EPO = secondary (the marrow is responding correctly to hypoxia).

PRACTICE

Q1. A patient with polycythemia has an increased risk of thrombosis. Which mechanism directly explains this risk?

- A. Increased blood viscosity slows flow and promotes clotting
- B. Decreased blood viscosity speeds flow and promotes clotting
- C. Increased plasma volume dilutes clotting factors
- D. Decreased platelet activity impairs clot formation

A is correct. A higher RBC count thickens the blood, slowing circulation and promoting clot formation. Viscosity *increases* (not decreases) in polycythemia, and platelet activity is normal or increased.

PRACTICE

Q2. Which finding suggests *secondary* rather than *primary* polycythemia?

- A. Elevated EPO from chronic hypoxia (e.g., COPD or high altitude)
- B. A JAK2 mutation with suppressed EPO
- C. Autonomous marrow overproduction of all cell lines
- D. Pruritus after a warm shower

A is correct. Secondary polycythemia is an appropriate marrow response to hypoxia, so EPO is *elevated*. The JAK2 mutation, suppressed EPO, multi-line overproduction, and post-shower pruritus all point to *primary* polycythemia vera.

When hemostasis fails, the patient **bleeds**. The lab signature tells you whether the problem is platelets (primary) or clotting factors (secondary).

Disorder	Defect	Lab signature
Hemophilia A	X-linked deficiency of factor VIII	↑ aPTT, normal PT, normal platelets
Hemophilia B	X-linked deficiency of factor IX	↑ aPTT, normal PT, normal platelets
Von Willebrand disease	Deficient/defective vWF → poor platelet adhesion	↑ aPTT, prolonged bleeding time, normal platelet count
Thrombocytopenia (ITP)	Autoimmune platelet destruction	↓ platelets, normal PT/aPTT, petechiae & purpura
Vitamin K deficiency	Impaired factors II, VII, IX, X	↑ PT > ↑ aPTT; responds to vitamin K

Bleeding pattern tells the level: Platelet/vWF problems → *superficial mucosal* bleeding — petechiae, easy bruising, nosebleeds, heavy menses, prolonged bleeding time (Sofia's case study = von Willebrand disease). **Clotting-factor problems** (hemophilia) → *deep* bleeding into joints and muscles, with a prolonged aPTT.

vWF does double duty: it bridges platelets to collagen (primary hemostasis) *and* stabilizes factor VIII. That is why von Willebrand disease prolongs the **bleeding time** *and* can mildly prolong the **aPTT**.

PRACTICE

Q1. A patient with thrombocytopenia experiences prolonged bleeding. Which mechanism explains this effect?

- A. Reduced platelet count impairs early plug formation
- B. Increased clotting factors accelerate fibrin formation

- C. Reduced vascular leakage limits platelet activation
- D. Increased erythrocyte count disrupts clot formation

A is correct. Too few platelets weaken primary hemostasis, so the initial plug forms poorly and bleeding is prolonged. The other options would not impair the early platelet plug.

PRACTICE

Q2. A 19-year-old has lifelong nosebleeds, easy bruising, heavy menses, petechiae, prolonged bleeding time, and a mildly prolonged aPTT with a normal platelet count. Which defect best explains this?

- A. Low von Willebrand factor activity impairing platelet adhesion
- B. Low GPIIb/IIIa activity impairing aggregation
- C. Low factor X activity impairing thrombin formation
- D. Low plasmin activity impairing fibrin breakdown

A is correct. vWF deficiency impairs platelet adhesion *and* destabilizes factor VIII, fitting the mucosal-bleeding pattern, prolonged bleeding time, and mildly prolonged aPTT with normal platelet count. C affects secondary hemostasis; D impairs clot breakdown, not formation.



Here the problem is pathologic **thrombosis** — clots forming where they should not.

Disorder	Mechanism	Clinical clue
DVT	Virchow's triad: stasis + endothelial injury + hypercoagulability	Unilateral leg swelling, warmth, pain; ↑ D-dimer
Pulmonary embolism	DVT embolizes to the pulmonary circulation	Sudden dyspnea, pleuritic chest pain, hypoxia; ↑ D-dimer
Factor V Leiden	Factor V resists inactivation by protein C	Inherited; recurrent VTE; family history
Antiphospholipid syndrome	Autoantibodies promote thrombosis	Recurrent miscarriage; arterial & venous clots
HIT	Antibodies activate platelets after heparin exposure	Platelet drop 5–10 days after heparin; paradoxical clotting

Virchow's triad — the master key: nearly every clotting scenario reduces to **stasis** (immobility, long flights), **endothelial injury** (trauma, surgery), or **hypercoagulability** (cancer, pregnancy, inherited thrombophilia). Immobility → venous stasis → DVT is the most tested example.

HIT paradox: heparin is given to *prevent* clots, but in HIT it triggers antibodies that *activate* platelets → the platelet count drops *and* the patient clots. Stop the heparin.

PRACTICE

Q1. A patient who has been immobile for several days develops thrombosis. Which mechanism best explains this condition?

A. Slowed circulation (venous stasis)

- B. Damaged vessel lining
- C. Reduced clot breakdown
- D. Increased platelet binding

A is correct. Immobility promotes venous stasis (one arm of Virchow's triad), slowing venous return and raising clot risk. Endothelial injury and platelet adhesion matter more in injury-related clotting.

PRACTICE

Q2. Which set of findings best fits a pulmonary embolism?

- A. Sudden dyspnea, pleuritic chest pain, hypoxia, and elevated D-dimer
- B. Petechiae, low platelets, and normal PT/aPTT
- C. Jaundice, elevated indirect bilirubin, and high reticulocytes
- D. Microcytic cells with low ferritin

A is correct. A PE classically presents with abrupt dyspnea, pleuritic pain, hypoxia, and a raised D-dimer. B is ITP, C is hemolytic anemia, and D is iron-deficiency anemia.

Disseminated intravascular coagulation (DIC) is the paradox where the patient **clots and bleeds at the same time** — and the capstone of the whole unit.

Pathophysiology

A trigger causes **systemic activation of coagulation** → widespread microthrombi form throughout small vessels → this **consumes platelets and clotting factors** faster than they can be replaced → severe **bleeding** follows.

Triggers: sepsis (most common), trauma, obstetric complications (placental abruption, amniotic fluid embolism), malignancy, and transfusion reactions.

Lab signature: ↑ PT, ↑ aPTT, ↓ platelets, ↓ fibrinogen, ↑↑ D-dimer. **Signs:** oozing from IV sites, hematuria, purpura, and organ dysfunction from microthrombi.

The lab-signature framework — four numbers solve most coagulopathies

Pattern	PT · aPTT · Plt · D-dimer	Most likely diagnosis
Isolated ↑ aPTT	nl · ↑ · nl · nl	Hemophilia, heparin, vWD
Isolated ↑ PT	↑ · nl · nl · nl	Warfarin, vitamin K deficiency, early liver disease
Both prolonged	↑ · ↑ · nl · nl	Liver disease, common-pathway defect
Low platelets only	nl · nl · ↓ · nl	ITP, drug-induced, marrow failure
Everything off	↑ · ↑ · ↓ · ↑↑	DIC
Isolated ↑ D-dimer	nl · nl · nl · ↑	DVT, PE

How to spot DIC instantly: it is the only entry where **everything is abnormal at once** — both clotting times prolonged, platelets and fibrinogen consumed (low),

and D-dimer sky-high — in a sick patient (sepsis, trauma, obstetric event) who is both bleeding and clotting.

PRACTICE

Q1 (Select all). A patient with sepsis develops oozing from IV sites, ↑PT, ↑aPTT, low platelets, low fibrinogen, and very high D-dimer. Which findings support DIC?

- A. Bleeding from multiple sites
- B. Both PT and aPTT are prolonged
- C. Platelet count is decreased
- D. Fibrinogen is elevated
- E. D-dimer is markedly elevated

A, B, C, and E are correct. DIC consumes factors and platelets (prolonged PT/aPTT, low platelets) while massively raising D-dimer, and bleeding appears at multiple sites. D is wrong — fibrinogen is *consumed*, so it falls, not rises.

PRACTICE

Q2. A lab panel shows an isolated prolonged PT with normal aPTT, platelets, and D-dimer. Which is most likely?

- A. Warfarin therapy or vitamin K deficiency
- B. Hemophilia A
- C. DIC
- D. Immune thrombocytopenia (ITP)

A is correct. An isolated long PT points to the extrinsic pathway — warfarin or vitamin K deficiency. Hemophilia prolongs the aPTT, DIC throws everything off, and ITP shows isolated low platelets.

Disorders of white cells affect **defense**. The two patterns to know are too few functional WBCs (leukopenia) and malignant overgrowth (leukemia).

Disorder	Mechanism	Consequence
Leukopenia	Reduced WBC number (chemo, radiation, marrow suppression)	Weak defense → frequent infections
Leukemia	Malignant cells crowd the marrow , reducing functional neutrophils	Infection (low functional WBCs), anemia, bleeding (crowded-out RBCs/platelets)
Pancytopenia	Marrow damage (radiation, aplastic anemia) drops all lines	Infection + anemia + bleeding together

Key idea — leukemia's infection paradox: the WBC *count* may look high, but the cells are abnormal and the marrow is so crowded it cannot make **functional neutrophils**. Fewer working neutrophils → infection risk — while crowding-out of RBC and platelet precursors adds anemia and bleeding.

Environmental link: bone-marrow toxins (benzene, certain chemotherapies) and radiation suppress production → anemia or full pancytopenia, depending on dose.

PRACTICE

Q1. A patient with leukopenia develops frequent infections. Which alteration explains this increased risk?

- A. Reduced cellular defense against invading organisms
- B. Impaired oxygen transport by erythrocytes
- C. Altered platelet function during vascular injury
- D. Disrupted plasma protein activity

A is correct. Fewer leukocytes mean less ability to recognize and eliminate pathogens. Oxygen transport, platelet function, and plasma proteins do not provide immune defense.

PRACTICE

Q2. A child with leukemia develops frequent infections. Which pathophysiologic change explains this finding?

- A. Marrow replacement reducing functional neutrophils
- B. Reduced immunoglobulin synthesis weakening humoral defense
- C. Decreased complement activity weakening opsonization
- D. Impaired cytokine signaling weakening cell recruitment

A is correct. Leukemic cells crowd the marrow and reduce production of functional neutrophils, the main driver of infection risk here. The other mechanisms can affect immunity but are not the primary problem in leukemia.

SECTION 3

Clinical Application

How these disorders present at the bedside and how nurses manage anemia, sickle cell crisis, DIC, transfusions, and pediatric hematology.

One deficit — reduced oxygen delivery — produces findings across every body system. Knowing the spread helps you assess and teach.

System	Findings
Cardiopulmonary	Tachycardia, palpitations, dyspnea on exertion, systolic flow murmur, widened pulse pressure
Neurologic	Headache, dizziness, syncope, poor concentration, paresthesias (B12)
Skin & mucosa	Pallor (conjunctiva, palms, nail beds), brittle nails, glossitis, cheilosis
GI	Anorexia, nausea, pica (iron deficiency)
Functional	Fatigue, cold intolerance, exercise intolerance — an early teaching target

Tie it together: tachycardia and the flow murmur are the heart moving thinner blood faster; pallor is reduced hemoglobin in the tissues; paresthesias flag **B12** specifically; pica and spoon nails flag **iron**. The assessment findings are the compensation mechanisms made visible.

PRACTICE

Q1. A patient with anemia develops fatigue and dyspnea. Which mechanism explains these findings?

- A. Low hemoglobin reduces oxygen delivery to tissues
- B. Blood loss reduces the volume of circulating plasma
- C. Hemolysis shortens the lifespan of red blood cells
- D. Marrow failure lowers production of red blood cells

A is correct. Anemia lowers oxygen-carrying capacity, decreasing oxygen delivery and causing fatigue and dyspnea. The other options are *causes* of anemia, not the direct reason for these symptoms.

PRACTICE

Q2. A patient with anemia has a reduced red blood cell count. Which change in blood properties contributes to altered circulation?

- A. Decreased viscosity causing faster blood flow
- B. Increased viscosity causing slower blood flow
- C. Increased osmotic pressure causing edema
- D. Decreased plasma volume causing hypovolemia

A is correct. Fewer red cells *lower* blood viscosity, allowing faster flow — one compensatory way to move oxygen. Viscosity does not rise in anemia, and osmotic pressure/plasma volume are governed by plasma proteins.

Crisis care reverses the triggers and treats the pain. Remember it as **hydrate, oxygenate, treat pain, watch the chest**.

- **Hydration** — IV fluids reverse the dehydration trigger and reduce sickling.
- **Oxygen** — supplemental O₂ if SpO₂ < 95% or symptomatic.
- **Pain control** — opioids around the clock; **do not undertreat**. Believe the patient's pain report.
- **Watch for acute chest syndrome** — fever + chest pain + new infiltrate is an emergency; exchange transfusion may be needed.
- **Long-term: hydroxyurea** raises fetal hemoglobin (HbF) and reduces crisis frequency; plus vaccines and folate supplementation.

Why hydroxyurea works: it boosts **HbF**, which does not sickle, diluting HbS and cutting the frequency of vaso-occlusive crises. Folate supports the high RBC turnover of chronic hemolysis; vaccines guard a functionally asplenic patient.

Each intervention undoes a trigger: fluids fix dehydration, oxygen fixes hypoxia, warmth avoids cold-induced sickling, and prompt infection treatment removes another precipitant. The management list *is* the trigger list, reversed.

PRACTICE

Q1. Which intervention most directly reverses a primary trigger of sickling during a vaso-occlusive crisis?

- A. IV hydration to correct dehydration
- B. Restricting fluids to reduce viscosity
- C. Applying cold packs to painful joints
- D. Withholding opioids to avoid sedation

A is correct. Dehydration promotes sickling, so IV fluids directly reverse it. Fluid restriction and cold worsen sickling, and undertreating pain is never appropriate in crisis.

PRACTICE

Q2. A patient in sickle cell crisis develops fever, chest pain, and a new pulmonary infiltrate. What does this most likely represent?

- A. Acute chest syndrome — a medical emergency
- B. A normal part of every crisis requiring no change
- C. Simple anxiety from pain
- D. A transfusion reaction

A is correct. Fever + chest pain + new infiltrate signals acute chest syndrome, an emergency that may require exchange transfusion. It is not a benign or expected finding.



DIC is always **secondary** to something else, so the cornerstone is to **find and reverse the cause** while supporting the patient.

- **Identify and reverse the trigger** — sepsis (antibiotics, source control), obstetric event, trauma, or malignancy.
- **Replace what is consumed** — **FFP** for clotting factors, **cryoprecipitate** for fibrinogen, **platelets** for thrombocytopenia, packed RBCs for blood loss.
- **Supportive ICU care** — hemodynamic stabilization, vasopressors as needed, organ support.
- **Anticoagulation** — controversial; sometimes used in chronic or thrombosis-predominant DIC.
- **Monitor** — trends in PT, aPTT, platelets, fibrinogen, and D-dimer guide therapy.

Core principle: you cannot win by only replacing products — if the trigger (e.g., sepsis) keeps activating coagulation, consumption continues. Treat the cause first; replace factors and platelets to support the patient in the meantime.

PRACTICE

Q1. What is the most important principle in managing DIC?

- A. Identify and treat the underlying trigger
- B. Give anticoagulation to every patient immediately
- C. Withhold all blood products
- D. Restrict fluids aggressively

A is correct. DIC is secondary; reversing the trigger (e.g., sepsis) stops the ongoing consumption. Universal anticoagulation is not standard, and replacement products are given as needed — not withheld.

PRACTICE

Q2. Which product correctly replaces fibrinogen in a patient with DIC?

- A. Cryoprecipitate
- B. Packed red blood cells
- C. Normal saline
- D. Albumin

A is correct. Cryoprecipitate is the concentrated source of fibrinogen. Packed RBCs replace red cells, saline is a crystalloid, and albumin supports oncotic pressure — none replaces fibrinogen.



Whenever a reaction is suspected, the first action is almost always the same: **stop the transfusion** and keep the line open with normal saline. The timing and mechanism then point to the type.

Reaction	Onset / mechanism	Action
Acute hemolytic	Within minutes — ABO incompatibility	Stop; saline; notify MD; send blood-bank workup
Febrile non-hemolytic	1–6 hr — recipient antibodies vs. donor WBCs	Stop; antipyretic; use leukoreduced products
Allergic / anaphylaxis	Minutes — donor plasma proteins	Stop; antihistamine; epinephrine if anaphylactic
TRALI	Within 6 hr — donor anti-leukocyte antibodies → pulmonary edema	Stop; respiratory support; ICU
TACO	Volume overload during/after transfusion	Slow/stop infusion; diuretics; oxygen

TRALI vs. TACO — both cause dyspnea: **TRALI** is an immune lung injury (anti-leukocyte antibodies) — treat with respiratory support. **TACO** is circulatory *overload* (too much volume, too fast) — treat with diuretics and slowing the infusion. Same symptom, opposite fluid logic.

The dreaded one: an **acute hemolytic reaction** from ABO incompatibility begins within minutes with fever, flank/back pain, and dark urine. Stop immediately — it can progress to shock and DIC.

PRACTICE

Q1. Minutes into a transfusion, a patient develops fever, flank pain, and dark urine. What is the priority action?

- A. Stop the transfusion immediately and infuse saline
- B. Slow the rate and continue

- C. Give an antipyretic and proceed
- D. Add more units to dilute the reaction

A is correct. These signs suggest an acute hemolytic reaction (ABO incompatibility). Stop the transfusion at once, keep the line open with saline, and notify the provider/blood bank. Continuing is dangerous.

PRACTICE

Q2. A patient develops dyspnea and pulmonary edema from volume overload during transfusion. Which reaction is this, and what is the treatment?

- A. TACO — slow/stop the infusion, give diuretics and oxygen
- B. TRALI — give more fluids rapidly
- C. Allergic reaction — give epinephrine first-line
- D. Febrile non-hemolytic — give antibiotics

A is correct. Transfusion-associated circulatory overload (TACO) is treated by slowing/stopping the infusion, diuresis, and oxygen. TRALI is an immune injury (not treated with more fluids), and the other reactions have different mechanisms.

Newborns and children have unique hematologic patterns that flow directly from the same physiology — mostly the EPO loop and RBC turnover.

Condition	Mechanism
Neonatal polycythemia (high RBCs at birth)	Fetal hypoxia in the low-oxygen womb drives high EPO → high RBC mass before birth
Anemia of prematurity	Immature kidneys make too little EPO after birth → under-stimulated marrow; neonatal RBCs also have shorter lifespans
Neonatal jaundice (< 24 hr)	Rapid breakdown of fetal RBCs → high unconjugated bilirubin
Hemolytic disease of the newborn	Maternal antibodies destroy fetal RBCs → destruction exceeds replacement → anemia
Hereditary spherocytosis	Non-deformable spherocytes trapped in the spleen → extravascular hemolysis, jaundice, gallstones

Same loop, different age: the fetus lives in a low-oxygen environment, so EPO runs high → polycythemia at birth. After birth, oxygen rises and EPO normally falls; in a **premature** infant the immature kidney under-produces EPO → anemia. Chronic hemolysis (spherocytosis) keeps reticulocytes high through that same EPO drive (Iris's case: elevated reticulocytes, marrow hyperplasia, spherocytes, high unconjugated bilirubin and urobilinogen, gallstone risk).

PRACTICE

Q1. A premature infant develops anemia during the first weeks of life. Which mechanism best explains this condition?

- A. Reduced erythropoietin from immature kidneys limits RBC production
- B. Maternal antibodies destroy circulating red blood cells
- C. Blood loss lowers the number of circulating red cells
- D. Bone marrow failure limits new red blood cell production

A is correct. In prematurity, immature kidneys produce less EPO, so the marrow is under-stimulated — the anemia of prematurity. Antibody-mediated destruction (B) is hemolytic disease of the newborn, a different mechanism.

PRACTICE

Q2. An 8-year-old has jaundice, elevated unconjugated bilirubin, increased urobilinogen, and numerous spherocytes on smear. Which change explains the premature red cell destruction?

- A. Loss of deformability traps cells in splenic sinusoids
- B. Loss of hemoglobin lowers tissue oxygen delivery
- C. Loss of ATP weakens membrane pump function
- D. Loss of membrane charge increases clumping

A is correct. Spherocytes are rigid and non-deformable, so the spleen traps and removes them (extravascular hemolysis). Lower hemoglobin contributes to anemia symptoms but does not explain splenic trapping.

EXAM

Lab-Signature Cheat Sheet

The highest-yield patterns — review these the night before.

The four coagulation numbers

Pattern (PT · aPTT · Plt · D-dimer)	Think
nl · ↑ · nl · nl	Hemophilia A/B, heparin, von Willebrand
↑ · nl · nl · nl	Warfarin, vitamin K deficiency, early liver disease
↑ · ↑ · nl · nl	Liver disease, common-pathway defect
nl · nl · ↓ · nl	ITP, drug-induced, marrow failure

↑ · ↑ · ↓ · ↑ ↑	DIC (everything off)
nl · nl · nl · ↑	DVT, PE

Anemia by size (MCV) + the discriminator

MCV	Type	One-clue discriminator
< 80 micro	Iron deficiency	Ferritin LOW , TIBC high , pica/spoon nails
< 80 micro	Anemia of chronic disease	Ferritin HIGH (hepcidin traps iron)
< 80 micro	Thalassemia	Normal iron studies, target cells, family history
> 100 macro	B12 deficiency	Neuro signs (paresthesias, ataxia)
> 100 macro	Folate deficiency	No neuro signs; alcohol, pregnancy, methotrexate
80–100 normo	Hemolytic	↑ retics, ↑ LDH, ↑ indirect bili, ↓ haptoglobin
80–100 normo	Aplastic / CKD	Pancytopenia (aplastic) or ↓ EPO (CKD)

Don't-miss facts

- **EPO trigger = low oxygen** (altitude, anemia, hemorrhage, lung disease) — kidney is the sensor.
- **CO** occupies the Hb site (carboxyhemoglobin); **methemoglobin** = iron oxidized to ferric (Fe^{3+}), can't bind O_2 .
- **Sickle cell** = glutamic acid → valine on beta-globin; HbS polymerizes when **deoxygenated**. Triggers: hypoxia, dehydration, infection, cold, acidosis, stress.
- **Polycythemia**: primary (PV) = JAK2, EPO **low**; secondary = hypoxia, EPO **high**. Both clot from hyperviscosity.
- **Bleeding (not enough glue)**: hemophilia (↑aPTT, deep joint/muscle bleeds), vWD (↑bleeding time + mild ↑aPTT, mucosal), ITP (low platelets, petechiae).

- **Clotting (too much glue):** Virchow's triad = stasis + endothelial injury + hypercoagulability. Immobility → DVT.
- **DIC** = clots *and* bleeds; trigger is usually **sepsis**; treat the trigger + replace (FFP, cryo, platelets).
- **Transfusion:** any suspected reaction → **STOP + saline**. Acute hemolytic = ABO, minutes. TRALI (lung injury, support) vs. TACO (overload, diuretics).
- **Liver disease** → low albumin → low oncotic pressure → edema; also ↑PT/aPTT (factors not made).
- **Pediatrics:** fetal hypoxia → polycythemia at birth; prematurity → low EPO → anemia; spherocytes → splenic (extravascular) hemolysis.

□ TERMS

Glossary

Quick definitions of the Week 8 vocabulary.

Erythropoietin (EPO)

Kidney hormone released in response to hypoxia; stimulates the marrow to make red blood cells.

Hematocrit (Hct)

Percentage of blood volume made up of red blood cells.

Hemoglobin (Hb)

Iron-containing protein in RBCs that carries oxygen; four globin chains, each with one heme group.

Reticulocyte

Immature red blood cell; a rising count signals active marrow production (e.g., after bleeding or hemolysis).

MCV (mean corpuscular volume)

Average RBC size; classifies anemia as micro- (<80), normo- (80–100), or macrocytic (>100).

Ferritin

Storage form of iron; low in iron deficiency, high in anemia of chronic disease.

TIBC (total iron-binding capacity)

Reflects transferrin available to carry iron; high in iron deficiency.

Hepcidin

Inflammatory hormone that traps iron in storage by degrading ferroportin; central to anemia of chronic inflammation.

Hemostasis

The process of stopping bleeding: vascular spasm, platelet plug (primary), coagulation (secondary), fibrinolysis.

Primary hemostasis

Platelet adhesion, activation, and aggregation forming the initial plug.

Secondary hemostasis

The clotting cascade converting fibrinogen to fibrin to stabilize the clot.

von Willebrand factor (vWF)

Bridges platelets to collagen and stabilizes factor VIII.

PT / INR

Measures the extrinsic + common pathway; monitors warfarin.

aPTT

Measures the intrinsic + common pathway; monitors heparin.

D-dimer

Fibrin breakdown product; elevated in DVT, PE, and DIC.

Thrombocytopenia

Low platelet count; causes petechiae, purpura, and prolonged bleeding time.

Hemolysis

Premature destruction of RBCs; raises indirect bilirubin and reticulocytes, lowers haptoglobin.

Spherocyte

Rigid, sphere-shaped RBC trapped and destroyed by the spleen (extravascular hemolysis).

Polycythemia

Excess red blood cells; raises viscosity and clot risk. Primary (JAK2/PV) or secondary (hypoxia).

DIC

Disseminated intravascular coagulation: simultaneous widespread clotting and bleeding from consumed factors and platelets.

Virchow's triad

The three roots of thrombosis: stasis, endothelial injury, hypercoagulability.

Carboxyhemoglobin

Hemoglobin bound to carbon monoxide, unable to carry oxygen.

Methemoglobinemia

Hemoglobin iron oxidized to the ferric (Fe^{3+}) state, unable to bind oxygen.

Pancytopenia

Reduction of all three cell lines (RBCs, WBCs, platelets), as in marrow failure or radiation.

Leukopenia

Low white blood cell count; increases infection risk.